NeuroShield: Incident Investigation

B.E. Information Science & Engineering | Academic Project Report

IEEE-style layout; not an IEEE publication or validated research result.

I. ABSTRACT

Matched: High packet rate (>=500/s), Forward / reverse packet imbalance, SYN without ACK, Elevated byte rate (>=100000/s). This is a transparent rule index, not a trained CICIDS classifier or proof of an attack.

DATA PROVENANCE: SYNTHETIC PRESENTATION SCENARIO. Confidence and attribution values are illustrative, not measured outputs.

II. INCIDENT METADATA

Incident: INC-5C9FC53D7F48

Observed (UTC): 2026-10-12T06:11:21.149545+00:00

Generated (UTC): 2026-10-12T09:21:52.056Z

Signal: Network flow

Source: Submitted flow / port 80

Severity: Critical

Risk score: 100

Response: Pending review

III. EVIDENCE INTEGRITY

SHA-256 of normalized incident metadata and explanation (not a hash of the original packet or uploaded file):

6401e08fca3c4b1aa836e0052a5e21c9ca9c922593788b520f2bd4b06c3656b8

IV. FEATURE ATTRIBUTION MATRIX

Factor / illustrative contribution (points)

High packet rate (>=500/s): +40 points

Forward / reverse packet imbalance: +30 points

SYN without ACK: +20 points

Elevated byte rate (>=100000/s): +10 points

Weights are local example contributions. They are not causal evidence and are not guaranteed to sum to 100.

V. MITIGATION HISTORY

2026-10-12T06:11:21.149545+00:00: Synthetic backend event.

2026-10-12T09:21:13.117Z: Status ’ Contained.

VI. LIMITATIONS AND REPRODUCIBILITY

No live enforcement action was executed. Model benchmark targets displayed in the application are not empirical results. Voice descriptors and OCR extraction do not establish deepfake or receipt authenticity.

A reproducible evaluation requires the model artifact, dataset version, train/test split, random seed, feature schema, explanation method, hardware, and measured metrics.